NEWTON SCHOOL

ALGEBRA 1 | ANSWER KEY

Rational Expressions: Simplify vs Solve

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Standard: CCSS.MATH.CONTENT.HSA-APR.D.6 & HSA-REI.A.2

Part A — SIMPLIFY

Problem 1

Simplify: $\left(y - \frac{2}{y-1}\right)\left(y + \frac{1}{y-2}\right)$

Combine Fractions | Factor | Cancel

Step 1. First factor: common denominator $y-1$.

$$y - \frac{2}{y-1} = \frac{y(y-1)-2}{y-1} = \frac{y^2-y-2}{y-1} = \frac{(y-2)(y+1)}{y-1}$$

Step 2. Second factor: common denominator $y-2$.

$$y + \frac{1}{y-2} = \frac{y(y-2)+1}{y-2} = \frac{y^2-2y+1}{y-2} = \frac{(y-1)^2}{y-2}$$

Step 3. Multiply and cancel $(y-1)$ and $(y-2)$: $\frac{(y-2)(y+1)}{y-1} \cdot \frac{(y-1)^2}{y-2} = (y+1)(y-1)$

$\boxed{y^2 - 1}$

Problem 2

Simplify: $\frac{x^2-9}{x^2+5x+6}$

Factor Difference of Squares | Factor Trinomial

Step 1. Factor numerator (difference of squares): $x^2 - 9 = (x-3)(x+3)$

Step 2. Factor denominator: $x^2 + 5x + 6 = (x+2)(x+3)$

Step 3. Cancel the common $(x+3)$: $\frac{(x-3)\,\cancel{(x+3)}}{(x+2)\,\cancel{(x+3)}}$

$\boxed{\frac{x-3}{x+2}}$

Problem 3

Simplify: $\dfrac{3}{x-2} - \dfrac{2}{x+1}$

Common Denominator   Distribute

Step 1. LCD is $(x-2)(x+1)$. $\dfrac{3(x+1)}{(x-2)(x+1)} - \dfrac{2(x-2)}{(x-2)(x+1)}$

Step 2. Combine and distribute numerator: $\dfrac{3x+3-(2x-4)}{(x-2)(x+1)} = \dfrac{3x+3-2x+4}{(x-2)(x+1)}$

Step 3. Simplify numerator: $3x-2x+3+4 = x+7$.

$$\boxed{\dfrac{x+7}{(x-2)(x+1)}}$$

Problem 4

Simplify: $\dfrac{x^2-4}{x^2-x-6} \div \dfrac{x^2+4x+4}{x^2-9}$

Division → Multiply by Reciprocal | Factor All

Step 1. Change division to multiplication by the reciprocal: $\dfrac{x^2-4}{x^2-x-6} \cdot \dfrac{x^2-9}{x^2+4x+4}$

Step 2. Factor each polynomial: $\dfrac{(x-2)(x+2)}{(x-3)(x+2)} \cdot \dfrac{(x-3)(x+3)}{(x+2)(x+2)}$

Step 3. Cancel $(x+2)$ (once) and $(x-3)$: $\dfrac{(x-2)\cancel{(x+2)}}{\cancel{(x-3)}\cancel{(x+2)}} \cdot \dfrac{\cancel{(x-3)}(x+3)}{(x+2)(x+2)} = \dfrac{(x-2)(x+3)}{(x+2)^2}$

$$\boxed{\dfrac{(x-2)(x+3)}{(x+2)^2}}$$

Part B — SOLVE

Problem 5

Solve: $\dfrac{x}{x-1} - \dfrac{2}{x+1} = \dfrac{4}{x^2-1}$

Clear Denominators | Quadratic | Check Extraneous

Step 1. Note $x^2-1=(x-1)(x+1)$. Multiply everything by $(x-1)(x+1)$:
$x(x+1)-2(x-1)=4$

Step 2. Expand and simplify: $x^2+x-2x+2=4 \;\Rightarrow\; x^2-x-2=0$

Step 3. Factor: $(x-2)(x+1)=0 \Rightarrow x=2$ or $x=-1$.

Step 4 — CHECK. $x=-1$ makes denominators $x+1$ and x^2-1 equal zero.

⚠ x = −1 is EXTRANEOUS — REJECT

$$\boxed{x=2}$$

Problem 6

Solve: $\frac{1}{x} + \frac{1}{x+3} = \frac{1}{2}$

Clear Denominators | Quadratic | Check Extraneous

Step 1. LCD = $2x(x+3)$. Multiply each term: $2(x+3) + 2x = x(x+3)$

Step 2. Expand: $2x + 6 + 2x = x^2 + 3x$, so $4x + 6 = x^2 + 3x$.

Step 3. Rearrange: $x^2 - x - 6 = 0$.

Step 4. Factor: $(x-3)(x+2) = 0 \Rightarrow x = 3$ or $x = -2$.

Step 5 — CHECK. Neither makes a denominator zero ($x \neq 0, -3$). Both valid.

$x = 3$ or $x = -2$

Problem 7 — Work-Rate Word Problem

> Two workers finish together in 6 hr. Faster worker finishes alone 5 hr sooner than slower. Find each worker's time alone.

Rate Equation | Quadratic | Reject Negative Time

Step 1. Let t = slower worker's hours. Then faster = $t-5$. Rates: $\frac{1}{t}$ (slower), $\frac{1}{t-5}$ (faster), together $\frac{1}{6}$.

Step 2. Equation: $\dfrac{1}{t}+\dfrac{1}{t-5}=\dfrac{1}{6}$

Step 3. Multiply by $6t(t-5)$: $6(t-5)+6t=t(t-5)$

Step 4. Expand: $6t-30+6t=t^2-5t \;\Rightarrow\; 12t-30=t^2-5t$.

Step 5. Rearrange: $t^2-17t+30=0$.

Step 6. Factor: $(t-15)(t-2)=0 \Rightarrow t=15$ or $t=2$.

Step 7 — CHECK. If $t=2$, faster $=2-5=-3$ hr (impossible).

⚠ t = 2 REJECTED (negative time)

Slower: 15 hr · Faster: 10 hr

Problem 8

> Solve: $\dfrac{2x}{x-3}=\dfrac{6}{x-3}+\dfrac{x}{2}$

Clear Denominators | Quadratic | Check Extraneous

Step 1. LCD = $2(x-3)$. Multiply every term: $4x=12+x(x-3)$

Step 2. Expand: $4x=12+x^2-3x$.

Step 3. Rearrange: $0=x^2-7x+12$.

Step 4. Factor: $(x-3)(x-4)=0 \Rightarrow x=3$ or $x=4$.

Step 5 — CHECK. $x=3$ makes denominator $x-3$ zero. **⚠ x = 3 is EXTRANEOUS — REJECT**

$x=4$